# Interview Prep – Project Proposal

## 1. Context

Job seekers (students and professionals) preparing for interviews face multiple challenges. Resources are scattered across blogs, YouTube videos, coaching platforms, and company websites. This creates confusion and consumes time, often leading to incomplete preparation.

There is a strong need for a **centralized, AI-driven solution** that helps candidates prepare efficiently with tailored content specific to their resume, company, and job role.

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## 3. Proposed Solution

**Interview Prep Web App** – A one-stop solution providing personalized interview preparation with the following:

- **Resume Upload:** AI extracts candidate skills and generates relevant technical questions.
- **Company & Job Role Input:** Candidate provides details to get tailored content.
- **AI-Generated Q&A:** Technical + HR + company-specific cultural/policy insights.2. Problem Statement

**Problem:** Interview preparation is fragmented, non-personalized, and time-consuming. Candidates struggle to find: - Relevant technical and HR questions. - Company-specific policies, culture, and updates. - Structured practice platforms with feedback. **Impact:** Candidates feel underprepared, lack confidence, and miss job opportunities.

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## - **Mock Interview Mode (Future Scope):** AI interviewer simulates real interview rounds via chat and video call with performance feedback.

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## 4. Product Details

### Features

1. **User Inputs**

2. Resume upload (PDF/DOC)

3. Company name

4. Job role

5. **AI Outputs**

6. Tailored technical questions based on skills & role.

7. HR questions based on resume and personality fit.
8. Company insights: work culture, policies, inclusivity, women-centric initiatives.
9. **UI Design**
10. **Clean layout** with gradient background (purple-pink theme).
11. **Two sections:** Input panel (resume, company, role) and Output panel (questions, insights).
12. **Cards & buttons** for structured readability.
13. **Future Additions**
14. Mock interview chat with real-time AI feedback.
15. Video conferencing mode with AI interviewer.
16. Analytics dashboard showing progress and confidence rating.

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## 5. Tech Stack

- **Frontend:** React (Vite), Tailwind CSS, shadcn/ui, lucide-react icons
- **Backend:** Node.js / Express (for API handling)
- **Database:** MongoDB (user profiles, resumes, generated Q&A)
- **AI Layer:** OpenAI / Gemini APIs for Q&A and company insights
- **Deployment:** Vercel (frontend), Render/Heroku (backend)

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## 6. Skills Required

- **Frontend Developer (2 members):** React, Tailwind, UI/UX design, API integration
- **Backend Developer (1 member):** Node.js, Express, MongoDB
- **AI/ML Integrator (1 member):** API usage (OpenAI/Gemini), prompt engineering, NLP
- **Project Manager & QA (1 member):** Documentation, timeline tracking, testing, deployment

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## 7. Team Task Division & Timeline (August – October)

### Week 1 (Aug 18 – Aug 24)

- Setup environment, GitHub repo, assign roles.
- Frontend skeleton: UI wireframe, routing.
- Backend setup: Node + Express boilerplate.

### Week 2 (Aug 25 – Aug 31)

- Resume upload functionality (file input + storage).
- Basic company & role input forms.
- Backend API stubs.

### Week 3 (Sept 1 – Sept 7)

- AI integration for generic Q&A.
- UI connection with AI output.
- Database schema design.

### Week 4 (Sept 8 – Sept 14)

- Resume parsing integration.
- Tailored technical questions generation.
- HR questions integration.

### Week 5 (Sept 15 – Sept 21)

- Company-specific insights (policies, culture, updates).
- Enhanced UI polish (colors, cards, gradients).

### Week 6 (Sept 22 – Sept 28)

- Testing & bug fixing (frontend + backend).
- Deployment of alpha version (frontend + backend).

### Week 7 (Oct 1 – Oct 7)

- AI refinement with better prompts.
- Feedback loop on generated questions.

### Week 8 (Oct 8 – Oct 14)

- User testing with dummy resumes.
- QA, UI/UX improvements.

### Week 9 (Oct 15 – Oct 21)

- Final deployment.
- Documentation.

### Week 10 (Oct 22 – Oct 31)

- Buffer week for fixes & future scope kickoff (Mock Interview Mode).

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## 8. Future Scope

1. **AI-Powered Mock Interviews**

2. Text chat simulation with real-time AI feedback.

3. Video call mode with speech recognition and evaluation.

4. **Performance Analytics**

5. Candidate readiness score.
6. Progress tracking over multiple sessions.
7. **Personalized Recommendations**
8. Courses, blogs, and resources for weak areas.
9. **Collaboration**
10. Peer-to-peer mock interviews within the platform.

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## 9. Expected Outcome

By the end of October, the team will deliver a functional **Interview Prep MVP** that:

- Accepts resumes, company names, and job roles.
- Generates personalized technical & HR interview questions.
- Provides company-specific policies and insights.
- Offers a clean, modern, and interactive UI for preparation.

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✅This sets the foundation for **AI-powered interview simulations** as a future expansion.